

Practice

Suppose X is a continuous random variable with pdf $f_X(x) = cx^2$ for some c and $0 \leq x \leq 1$

a) Find the value of c

$\Rightarrow$ We need the total probability of 1

$$\int_0^1 f_X(x) dx = \int_0^1 cx^2 dx = c \left[\frac{x^3}{3} \right]_0^1 = c \left[\frac{1}{3} \right] = \frac{1}{3}c$$

$\Rightarrow$ We need $1 = \frac{1}{3}c$, so $c = \frac{3}{8}$

b) Find the cdf of X

$$\int_0^x f_X(t) dt = \int_0^x \frac{3}{8} t^2 dt = \frac{3}{8} \left[\frac{t^3}{3} \right] = \frac{1}{8} x^3 \Rightarrow F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} x^3 & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

c) Find $P(X > 1)$

2 ways $\Rightarrow P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1) = 1 - \frac{1}{8}(1)^3 = \frac{7}{8}$

$$\Rightarrow P(X > 1) = P(1 < X \leq 2) = \int_1^2 \frac{3}{8} x^2 dx = \frac{7}{8}$$

d) Find the 75th percentile of X (0.75 quantile)

$$\Rightarrow \frac{3}{4} = F_X(x) = \frac{1}{8} x^3$$

$$\Rightarrow \frac{24}{4} = x^3$$

$$\Rightarrow \sqrt[3]{\frac{24}{4}} = x$$

CDF Technique

If X is a CRV with pdf f_X . Suppose Y is a transformation of X . We can obtain p_Y by obtaining the cdf of $Y = g(X)$

Example

Suppose X is a random variable with $f_X(x) = \frac{1}{2}$ for $-1 < x < 1$. Find the pdf of $Y = X^2$

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} < X < \sqrt{y}) = P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y}) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$\text{Then } f_Y(y) = \frac{d}{dy} [F_Y(y)] = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) - \frac{1}{2\sqrt{y}} f_X(-\sqrt{y}) = \frac{1}{2} \frac{1}{2\sqrt{y}} + \frac{1}{2} \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}, \text{ where } S_Y = \{y : 0 < y < 1\}$$

Jacobian Method

* For one-to-one transformations

* If X is a CRV with pdf $f_X(x)$ and support S_X . Let $Y = g(X)$, where g is a one-to-one differentiable function on S_X . Then the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| \text{ for } y \in S_Y$$

where the support of Y is the set $S_Y = \{y = g(x) : x \in S_X\}$

* $\frac{dx}{dy} = \frac{d[g^{-1}(y)]}{dy}$ as the Jacobian (J) of the transformation

Example

Suppose X is a random variable with pdf $f_X(x) = 2x$ for $0 < x < 1$. Find the pdf of $Y = X^2$ using the Jacobian method.

$\Rightarrow$ Note that $g(x) = x^2$ is one-to-one over $(0, 1)$, where $g^{-1}(y) = \sqrt{y}$

$$\Rightarrow \frac{dx}{dy} = \frac{d}{dy} [g^{-1}(y)] = \frac{d}{dy} [\sqrt{y}] = \frac{1}{2\sqrt{y}}$$

$\Rightarrow$ By the Jacobian method theorem we have $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right| = 2(\sqrt{y}) \left(\frac{1}{2\sqrt{y}} \right) = 1$, where $S_Y = (0, 1)$